Report on

“Tutorials”

For

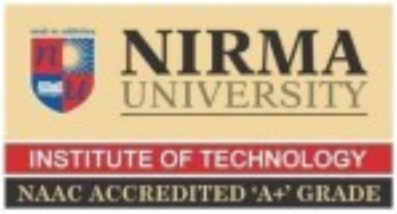
Software Project Management - 4CS504ME25 B.

Tech. Semester IV

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Aim

The main aim is to create a Work Breakdown Structure (WBS) for the Next-Generation Hospital Management System (HMS) project. It breaks the six main modules and other project work from T1 to T3 into smaller tasks and activities. This will help us organize the work, plan the project properly, and keep track of our progress.

Objective

- Divide the HMS project into the main deliverables based on the scope given in T1.
- Break each main deliverable into smaller parts, work packages, and tasks.
- Show the project work in a four-level WBS structure.
- Assign each main deliverable to the team member responsible for it, based on the roles given in T3.
- Put the complete WBS into one table so it is easy to understand and track.

1. Introduction

A Work Breakdown Structure (WBS) helps divide a project into smaller and easier-to-manage tasks. It starts with the main deliverables and then breaks them down into smaller parts and activities. A WBS is an important project management tool because it helps with planning the schedule, estimating the work, and assigning tasks to team members.

This report creates a WBS for the Next-Generation HMS project. It uses the six modules and project requirements from T1 , the sprint plan from T2 and the milestones and team roles from T3. The project is divided into four levels: major deliverables, sub-deliverables, work packages, and activities.

2. WBS Levels

The decomposition in this report follows four levels, each one step smaller than the one above it.

- **Level 1 - Major Deliverable:** The main outputs of the project, such as a completed HMS module or a project management task.
- **Level 2 - Sub-deliverable:** The main parts of a deliverable, such as backend development,

frontend development, or testing.

- **Level 3 - Work Package:** A group of related tasks that can be given to one team member or a small group to complete during a sprint.
- **Level 4 - Activity:** The small, specific tasks that need to be completed as part of a work package.

3. Hierarchical WBS

The HMS project is decomposed below into nine major deliverables. The first two cover project management and system design, the next six correspond to the modules defined in T1, and the last covers integration, testing, and deployment.

1. Project Management & Planning

- **1.1 Project Initiation**
    - **1.1.1 Team Formation**
        - 1.1.1.1 Define team roles and responsibilities
    - **1.1.2 Requirement Gathering**
        - 1.1.2.1 Collect system requirements
        - 1.1.2.2 Prepare the SRS document
- **1.2 Sprint Planning**
    - **1.2.1 Sprint Planning**
        - 1.2.1.1 Prepare sprint backlogs
        - 1.2.1.2 Set sprint goals

2. System Design & Architecture

- **2.1 Database Design**
    - **2.1.1 Database Schema**
        - 2.1.1.1 Design the ER diagram
        - 2.1.1.2 Finalize the database tables
- **2.2 Application Architecture**
    - **2.2.1 Backend and Frontend Design**
        - 2.2.1.1 Define API and UI structure

        - 2.2.1.2 Set up authentication and access control

3. Patient Registration & Management

- **3.1 Backend Development**
    - **3.1.1 Patient APIs**
        - 3.1.1.1 Develop CRUD functions (basically for patents)
- **3.2 Frontend Development**
    - **3.2.1 Patient UI**
        - 3.2.1.1 Build patient registration and record screens
- **3.3 Testing**
    - **3.3.1 Module Testing**
        - 3.3.1.1 Test of functions implemented on ly for patents

4. Doctor & Department Management

- **4.1 Backend Development**
    - **4.1.1 Doctor & Department APIs**
        - 4.1.1.1 Develop doctor and department functions
- **4.2 Frontend Development**
    - **4.2.1 Management UI**
        - 4.2.1.1 Build doctor and department screens
- **4.3 Testing**
    - **4.3.1 Functional Testing**
        - 4.3.1.1 Test doctor schedules and department functions

5. Appointment Booking & Scheduling

- **5.1 Backend Development**
    - **5.1.1 Booking APIs**
        - 5.1.1.1 Develop appointment booking functions
- **5.2 Frontend Development**
    - **5.2.1 Booking UI**
        - 5.2.1.1 Build the appointment booking screen
- **5.3 Testing**
    - **5.3.1 Integration Testing**

        - 5.3.1.1 Test appointment booking with patient and doctor modules

6. Pharmacy

- **6.1 Backend Development**
    - **6.1.1 Stock APIs**
        - 6.1.1.1 Develop medicine stock functions
- **6.2 Frontend Development**
    - **6.2.1 Pharmacy UI**
        - 6.2.1.1 Build stock and medicine screens
- **6.3 Testing**
    - **6.3.1 Stock Testing**
        - 6.3.1.1 Test stock updates and dispensing

7. Billing & Invoice

- **7.1 Backend Development**
    - **7.1.1 Billing APIs**
        - 7.1.1.1 Develop invoice generation functions
- **7.2 Frontend Development**
    - **7.2.1 Billing UI**
        - 7.2.1.1 Build invoice and payment screens
- **7.3 Testing**
    - **7.3.1 Calculation Testing**
        - 7.3.1.1 Test invoice calculations

8. Admin Dashboard & Reports

- **8.1 Backend Development**
    - **8.1.1 Reporting APIs**
        - 8.1.1.1 Develop basic reporting functions
- **8.2 Frontend Development**
    - **8.2.1 Dashboard UI**
        - 8.2.1.1 Build the admin dashboard
- **8.3 Testing**
    - **8.3.1 Report Testing**

        - 8.3.1.1 Check report accuracy

9. Integration, Testing & Deployment

- **9.1 System Integration**
    - **9.1.1 Integration**
        - 9.1.1.1 Combine all six modules
- **9.2 User Acceptance Testing**
    - **9.2.1 UAT**
        - 9.2.1.1 Conduct UAT using sample hospital scenarios
- **9.3 Deployment & Documentation**
    - **9.3.1 Deployment**
        - 9.3.1.1 Deploy the system
    - **9.3.2 Final Documentation**
        - 9.3.2.1 Finalize project documents
        - 9.3.2.2 Prepare the final demo

4. WBS Summary Table

The table below lists the nine major deliverables with their assigned owner, cross-referenced against the team roles defined in T3.

WBS	Major Deliverable	Primary Owner
1	Project Management & Planning	Both
2	System Design & Architecture	Both
3	Patient Registration & Management Module	Urva (Backend Lead)
4	Doctor & Department Management Module	Rakshit (Frontend Lead)
5	Appointment Booking & Scheduling Module	Rakshit
6	Pharmacy / Medicine Stock Module	Rakshit
7	Billing & Invoice Module	Urva

8	Admin Dashboard & Reports Module	Urva
9	Integration, Testing & Deployment	Both

5. Conclusion

This WBS basically divides the project into the task which is basically easily managable, also spring planning and assignment.